Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/30/2026
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Thermophysical Properties and Transport Phenomena Models and Experiments on Photorefractive Materials in Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	07/30/2024	**End Date:**	01/29/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Hanson, Ian	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/29/2027 per F. Hernandez/BPS (Ed., 4/9/26).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K1893		
Performance Goal No.:			
Performance Goal Text:			
	The objective of the proposed work is to advance the fundamental understanding of the origins of photorefractivity in certain photorefractive crystals and the manufacturing processes needed to apply this understanding to new devices here on Earth. The proposed program will consist of four major elements: continued development of novel non-contact methods for measuring thermophysical properties, including using electrostatic levitation in reduced gravity; application of these methods to materials of interest to industry and the international scientific community; application of the measured properties to produce accurate models of fluid flow and heat and mass transfer in the levitated samples; and application of the measured properties and models to test theories about the effect of processing on microstructure and material characteristics of a family of nonlinear optical crystals. The proposed elements primarily contribute to the research emphasis:		

Task Description:	1: thermophysical property measurements, with supporting contributions to research emphasis; 2: materials processes affecting microstructure, composition, and the resultant material characteristics. The proposed elements vary in scope from the development of methods (which are broadly applicable to measurement of thermophysical properties such as density and viscosity), to measurement of the properties of samples proposed by our team and other investigators in MaterialsLab, to models for specific experiments. Through the insight into the transport phenomena inside the samples, the experiment-specific models enable experiments that would otherwise be impossible. Similar models are currently in use for model-enabled experiments on the International Space Station (ISS). The results of the model-enabled experiments will advance the scientific understanding of the effect of processing on the properties of non-linear optical crystals, moving them closer to commercial production and sale here on Earth. This project will perform reduced gravity experiments using the Japanese Space Agency (JAXA) Electrostatic Levitation Furnace (ELF) aboard the International Space Station. These experiments will achieve precise measurements of the thermophysical properties of molten sillenites, specifically the density and thermal expansion coefficient as a function of temperature and time. The dependence on time is a test of a key hypothesis of the project: that the liquid experiences changes with sluggish kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Dr. Gwendolyn Bracker joined Worcester Polytechnic Institute (WPI) in October 2024 as an Assistant Research Professor. She continues to collaborate on this project, now under support from the subject grant. • Graduate Student Eric Klein and Principal Investigator (PI) Hyers submitted the Science Requirements Document for this project for review. • Mr. Klein and Prof. Hyers visited the NASA Marshall Space Flight Center (MSFC) Electrostatic Levitation (ESL) Lab August 5-9, 2024. In collaboration with ESL staff, they developed a process for fabricating flight samples rapidly. ~60 flight samples of Bismuth Sillenite (Bi12SiO20, BSO) and Bismuth Germanate (Bi12GeO20, BGO) were prepared. • The samples were tested at MSFC with a shock test expected to exceed the requirements for flight. The vast majority, but not all, of the samples passed. • The samples were sent to the Japanese Space Agency (JAXA) in December 2024 for qualification testing. • JAXA identified a previously known issue with thermal shock and the requirements for melting the sample. Revisions to the testing plan for qualification of the samples are ongoing. • The Science Requirements Review for this experiment has been postponed pending resolution of the issues with qualification of the flight samples. • Profs. Bracker and Hyers are both co-organizers of the symposium “Materials Research in Reduced Gravity”, to be held at The Minerals, Metals & Materials (TMS) Annual Meeting in March 2026. • Prof. Hyers is a co-organizer of the symposium “Z02: Electrochemical / Materials Processing in Space Engineering 2” at the 248th Electrochemical Society Meeting, Chicago, IL, October 12-16, 2025.
Bibliography Type:	Description: (Last Updated: 03/30/2026)